Experiment Number: 1

Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context

Aim:To simulate and compare the basic resource-allocation methods used in FDMA, TDMA and CDMA, and examine how theirassumed user capacity and interference behaviour differ.

Objective 2: CDMA SIR analysis:
To analyze how the estimated CDMA signal-to-interference ratio changes as the number of simultaneous users increases.

CODE:

```
% Experiment 1: CDMA SIR Analysis

clear;
clc;
close all;

% Parameters
processingGain = 64;
targetSIRdB = 7;

% Number of simultaneous users
K = 2:30;

% Calculate SIR
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);

% Display selected results
fprintf('CDMA SIR ANALYSIS\n');
fprintf(' ---------------- \n');
fprintf('Users\tSIR (dB)\n');

for k = [2 4 8 10 16 20 30]
   sir = 10 * log10(processingGain / (k - 1));
   fprintf('%d\t%.2f\n', k, sir);
end
```

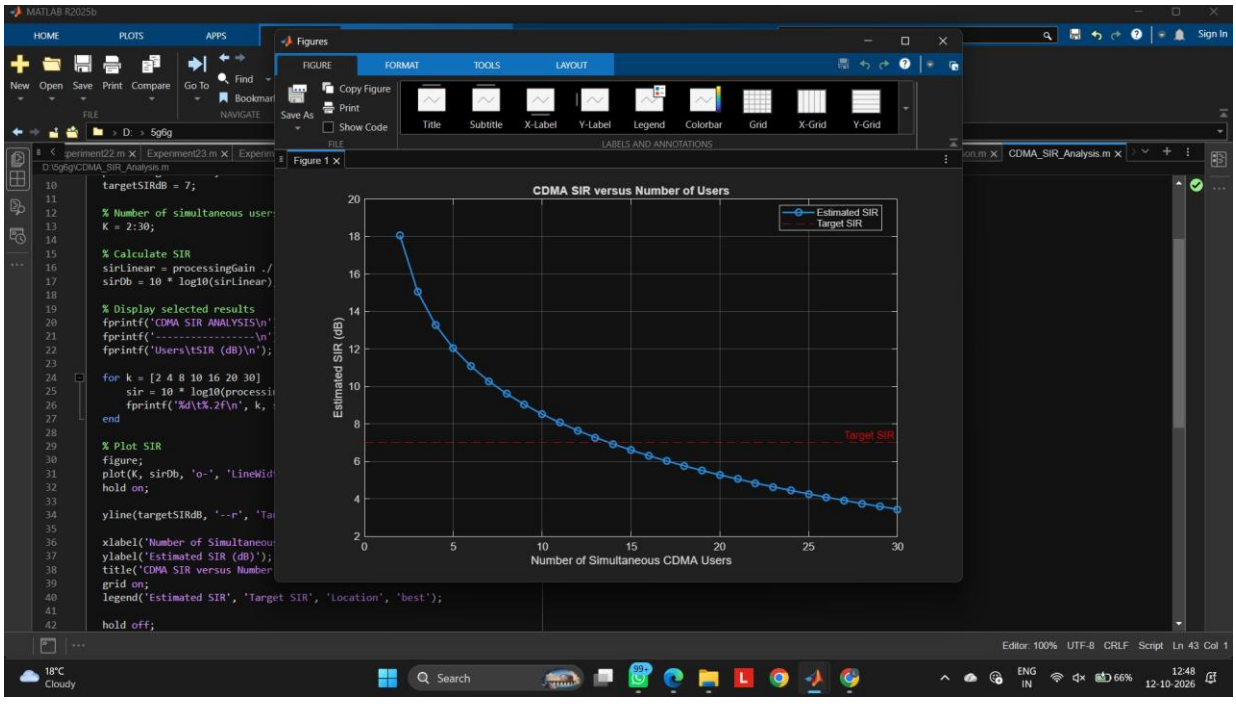
```
% Plot SIR
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;

yline(targetSIRdB, '--r', 'Target SIR');

xlabel('Number of Simultaneous CDMA Users');
ylabel('Estimated SIR (dB)');
title('CDMA SIR versus Number of Users');
grid on;
legend('Estimated SIR', 'Target SIR', 'Location', 'best');

hold off;
```

OUTPUT:



Performance Evaluation Rubric

Assessment Criterion	4 — Strong	3 — Meets	2 — Partial	1 — Limited
Objective 2: CDMA SIR analysis	Correct SIR conversion, calculation and plot; target line shown; trend explained.	Plot and calculations mostly correct, with minor errors or incomplete interpretation.	Plot or calculations contain errors; trend only partly explained.	Program or plot missing/incorrect, or trend misinterpreted.
Parameter study and recorded results	Changes relevant parameters, reruns applicable programs, records outputs and explains effects.	Provides rerun results and comparison, but one change or explanation is incomplete.	Makes limited changes or records results without clear comparison.	No meaningful parameter study or usable results record.
Analysis and interpretation	Answers accurately, uses results as evidence and distinguishes model behavior from real-system behavior.	Most answers sound and refer to results, with some gaps in explanation.	Answers partly correct but mainly describe outputs without explaining them.	Analysis missing, unsupported or substantially incorrect.
Engineering judgment and conclusion	Gives a clear conclusion, identifies assumptions and limitations, and avoids unsupported performance rankings.	Conclusion is reasonable but omits a relevant assumption, limitation or implication.	Conclusion mostly repeats results, with little discussion of limitations.	Conclusion missing or draws unsupported conclusions.
Report organization and evidence	Includes code, tables, required figures, parameter values and clear captions.	Required evidence present, with minor organization or labeling issues.	Some evidence missing or difficult to follow.	Report substantially incomplete or unclear.
Individual explanation / viva	Clearly explains own work, formulas and results.	Explains most aspects, with minor gaps.	Gives partial explanations but struggles to connect code and results.	Cannot explain key parts of the submitted work.